

Git Branching Strategy

→ A set of rule/conventions which defines the workflow

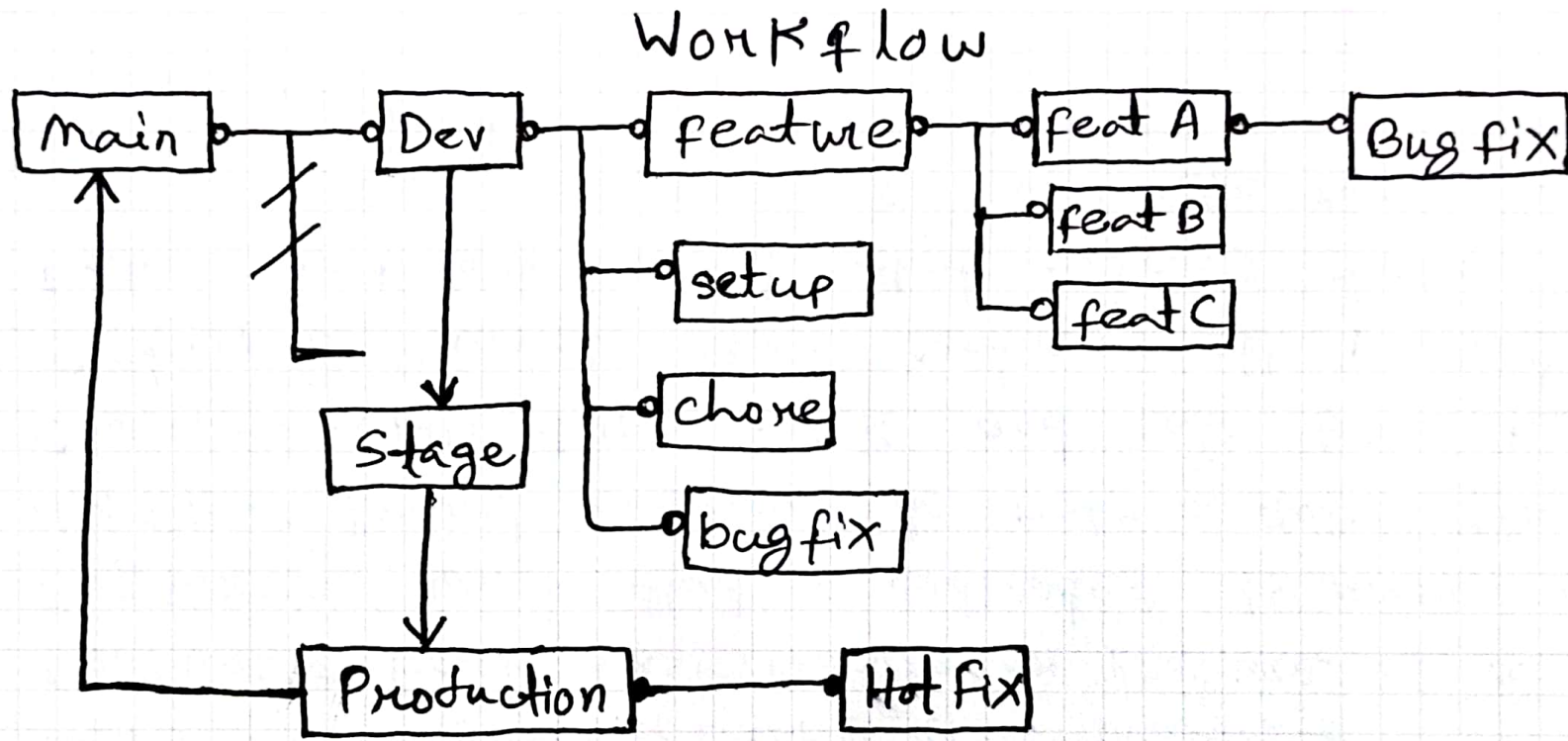
Typical git branch names

- Main / Master → Contains the ~~last~~ latest stable code
- Prod / Production → Dedicate branch that contains prod code
- Staging / Stage → Branch that is tested in production like
- Dev / Development → Branch that contains finalize codes from dev
- Feature → Each task, module or feature of a project
- Bug Fix → Branch that is dedicated to address a single bug
- Hot Fix → Branch dedicated to make a quick fix
- Chore → Util type task and code goes here
- Setup → Branch that contains all project setup file
- Release → ~~Branch~~ Branch that contains stable release code as well as release note and bug list

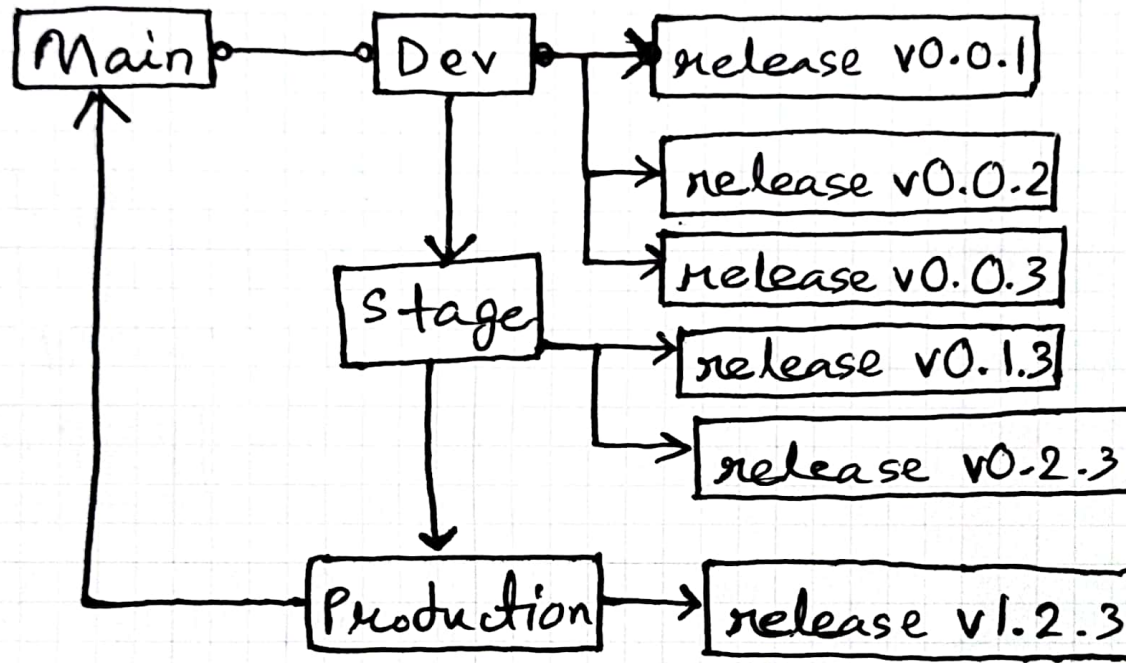
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Work flow



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Type of Software Based on Build & Deploy

Interpreted

Python, JS,
Ruby, PHP
FastAPI, Django
Express.js, laravel

Compiled

C, C++
Go, Rust
Bery

Bytecode

Java, C#
Spring Boot
.NET

Bundled

JS, React,
Vue.js,
Angular

Interpreted: They are interpreted at runtime and so not require build.
Can be ran directly after resolving dependencies.

Compiled: Require compilation before deployment. Compilation creates a standalone executable.

Bytecode: Compilation into an intermediate language, which is then interpreted or compiled by the runtime environment. JVM (Java), .NET runtime for example

Bundled: Require bundling code assets before deployment

Interpreted:

- Usually project is contained inside an isolated environment
- All dependency is resolved in that isolated env
- Usually have package manager to resolve dependency
- Can be executed directly
- Can be executed with web servers (Apache, Nginx)

Compiled:

- Requires build environment
- Build creates executable binaries (artifacts)
- Build files are moved to deployment server
- Executed in deployment server
- Deployment server needs to have runtime dependencies resolved

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Byte code:

- Requires build environment
- Build steps create an intermediary file; JAR, WAR, DLL for example
- File is transferred to deployment server
- Server requires appropriate runtime environment to execute code

Bundle:

- Package manager resolve dependency
- Build step bundle/minify code assets
- Deployment can be done using CDN or web server